

Going beyond aggregates: A dynamic rating system for NFL pass rushers and blockers

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Background

Currently, evaluation metrics for pass blockers and rusher consist of the following:

1. Pass block win rate
2. Aggregated statistics (sacks, hurries, etc.)
3. Film playback

We aim to apply an ELO model to dynamically update player ratings throughout a season, dependent on matchup outcomes

ELO

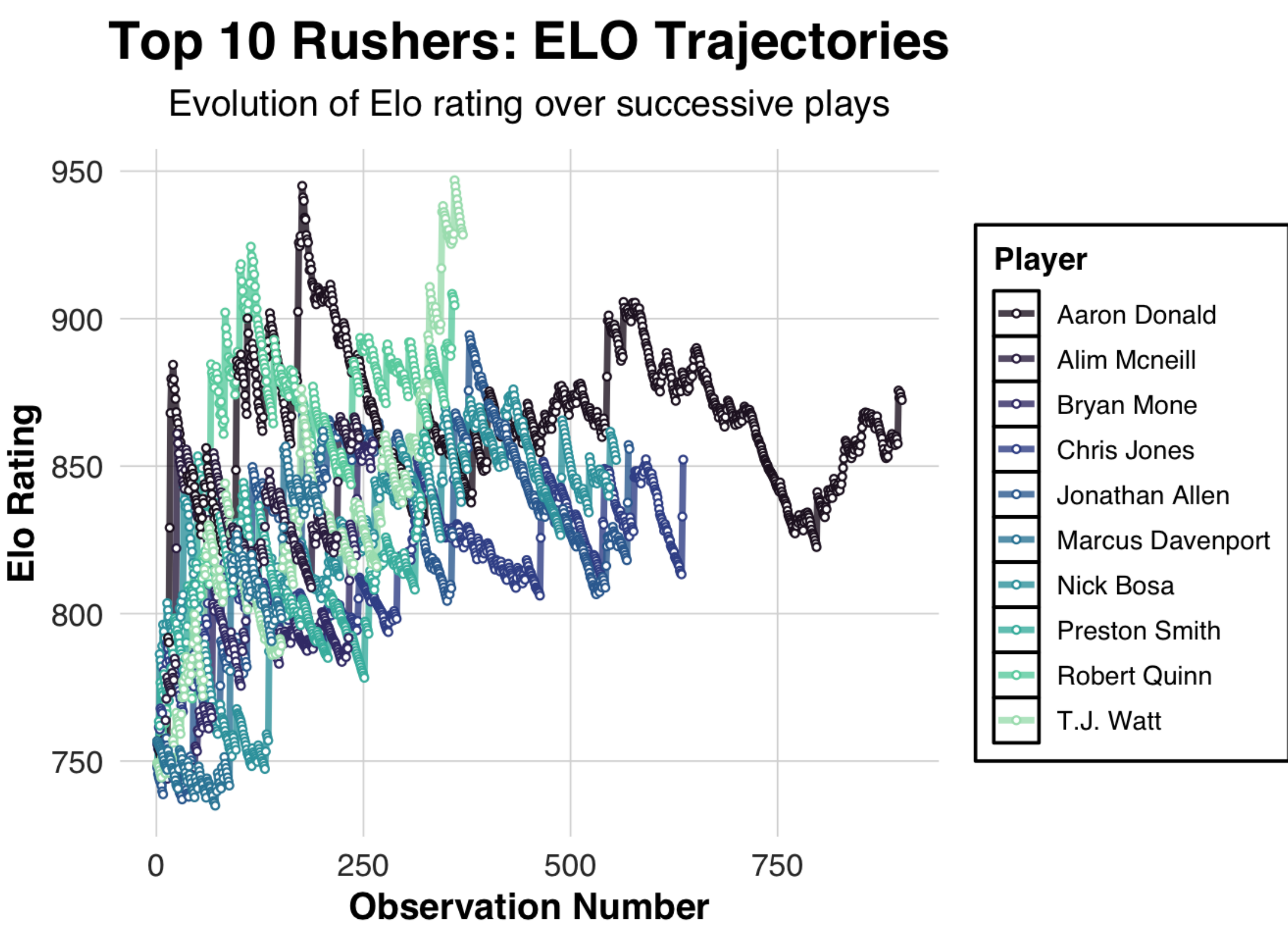
We utilize the ELO rating model to dynamically update player skill levels based on equation 1:

$$R'_A = R_A + K(S_A - E_A)$$

Where R'_A is the new player A ELO, R_A is the original player A ELO, K is the learning rate parameter, S_A is the event outcome, and E_A is the expected event outcome

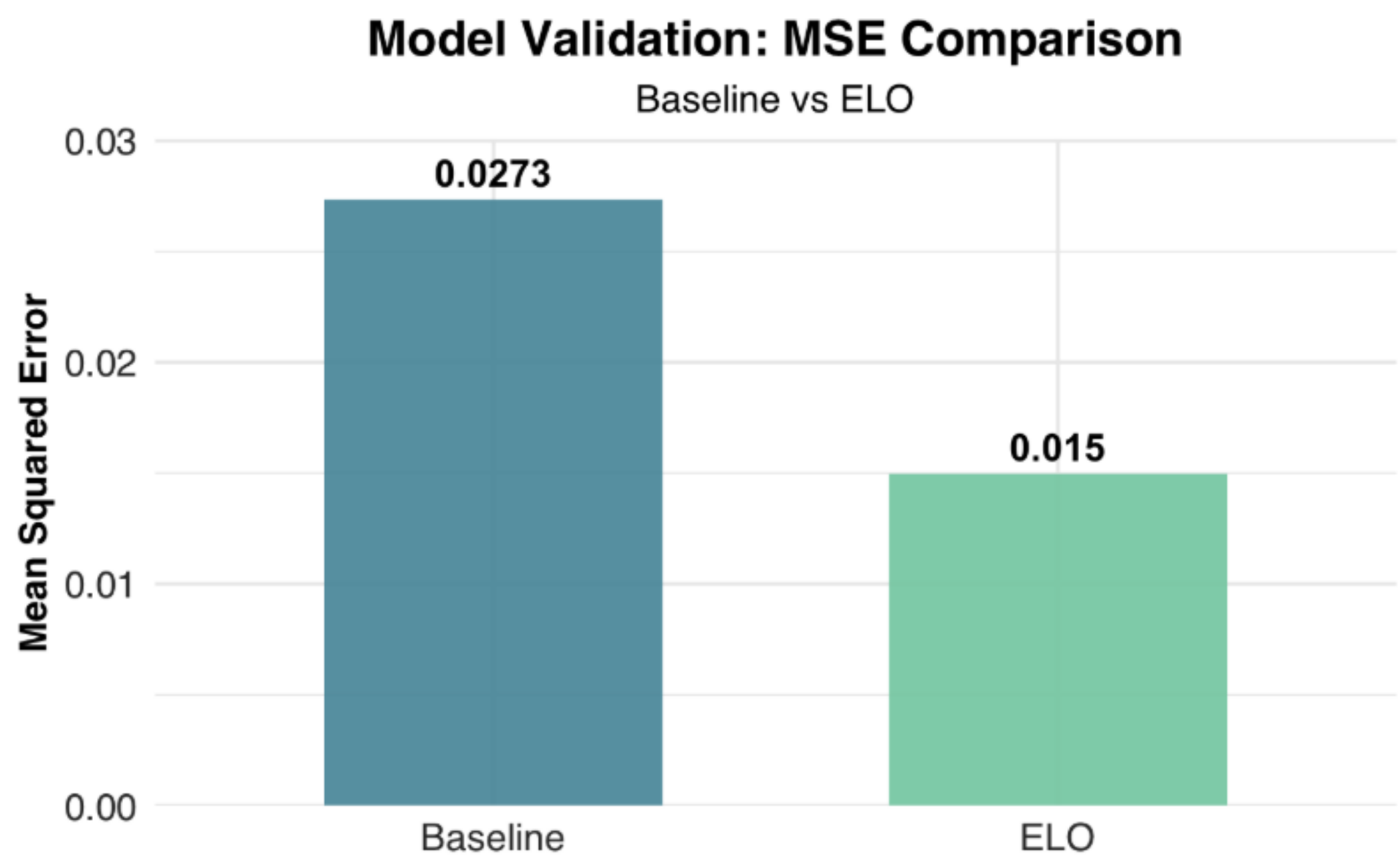
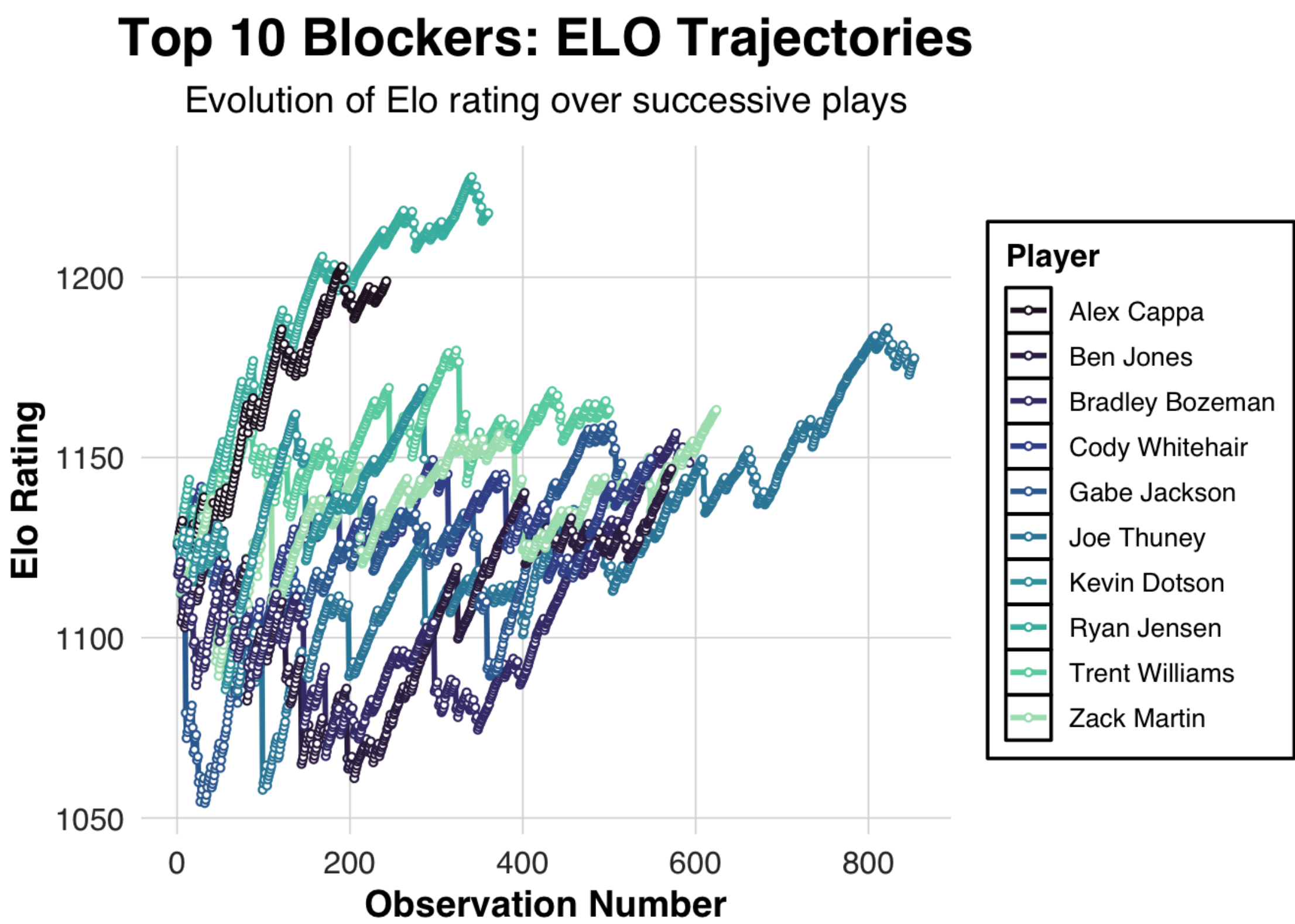
Results

We apply our model and generate the trajectories of the top 10 players on each side of the ball based on final ending ELO value. We see an increase in ELO value with stabilization at high observation numbers.



We validate with a baseline predictor of each player's average win rate across all interactions. We see a 45% improvement by predicting with player ELO differences, as shown in equation 3, where our scaling parameter is chosen as 319.

$$P(A \text{ wins}) = \frac{1}{1 + 10^{(R_B - R_A)/\alpha}}$$



Discussion

With ELO we see:

- Intuitive head-to-head comparisons — Elo naturally models matchups, making it easy to compare players directly
- Opponent-adjusted ratings — performance against stronger opponents carries more weight than against weaker ones
- Adaptive over time — ratings update sequentially, capturing momentum and mid-season changes in player quality
- Efficient with limited data — Elo produces stable estimates even when sample sizes are small (missed time due to injury, etc.)

Our current model only evaluates linemen on the pass block and rush. Evaluation on rushing plays would offer more complete insight on player ability. Additionally, further work can build off our model with multiple seasons of data as well as scaling results for greater interpretability.

Methods

1. Data preparation

Data for the '21-22 season acquired from Hudl was cleaned by filtering for pass plays and grouping by play to add all linemen and quarterback freeze frames for each play

2. Outcome identification

We use Euclidean distance from the QB to identify simple beats, with a weighing of 0.1. We then use tackles on the QB that occur before the pass and behind the line of scrimmage to identify hits, with a weighing of 0.2. Sacks have been pre-identified in the data and are weighed as 1.0

3. ELO updates

We feed each interaction to our multinomial ELO model sequentially. We also utilize an adaptive learning rate, as shown below, to linearly decrease the learning rate after observations $> n_{\text{prov}}$ and to set it to a minimum after n_{decay}

$$K(n) = \begin{cases} K_{\text{start}}, & \text{if } n \leq n_{\text{prov}} \\ K_{\text{start}} + (K_{\text{min}} - K_{\text{start}}) \cdot \frac{n - n_{\text{prov}}}{n_{\text{decay}} - n_{\text{prov}}}, & \text{if } n_{\text{prov}} < n \leq n_{\text{decay}} \\ K_{\text{min}}, & \text{if } n > n_{\text{decay}} \end{cases}$$

Acknowledgements

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